# I. Heat transfer

## Symbology

| Symbol | Quantity | Unit |
|---|---|---|
| $\lambda$ , $k$ | Thermal conductivity | $\mathrm{W/(m \cdot K)}$ |
| $\dot{q}$ | Heat flux | $\mathrm{W/m^2}$ |
| $\dot{Q}$ | Heat flow | W |
| $\alpha$ | Heat transfer coefficient | $\mathrm{W/(m^2 \cdot K)}$ |
| $a$ | Thermal diffusivity | $\mathrm{m^2/s}$ |
| $\nu$ | Kinematic viscosity | $\mathrm{m^2/s}$ |

## 1. Types of heat transfer

### 1.1 Heat conduction

Occurs with all materials when there is a temperature gradient.

**Steady-state conduction (1st Fourier's law)**:

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x}$$

**Transient conduction (2nd Fourier's law 1D)**:

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \frac{\partial^2 T}{\partial x^2}$$

### 1.2 Convective heat transfer

This occurs in moved fluids.

**Newton's approach**:

$$\dot{q} = \alpha \left(T_1 - T_2\right) = \frac{\lambda_{\text{fluid}}}{\delta_{\text{th}}} \left(T_1 - T_2\right)$$

### 1.3 Radiation

Bodies with a temperature above 0 K emit thermal radiation.

## 2. Heat conduction and conductivity

### 2.1 Heat flux

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x} = k\Delta T = \frac{\lambda}{\delta}\Delta T$$

Procedure:

$$\vec{\dot{q}} \int_0^{\delta} dx = -\lambda \int_{T_1}^{T_2} dT \Longrightarrow \dot{q} = \frac{\lambda}{\delta}\left(T_1 - T_2\right)$$

### 2.2 Heat flow

$$\dot{Q} = \frac{\Delta Q}{\Delta t} = \dot{q}A = kA\Delta T = \frac{\lambda}{\delta} A \Delta T$$

### 2.3 Thermal conduction $k^{-1}$ and Thermal conductivity $k$

$$k_{\text{tot}}^{-1} = \sum_{i=1}^{n} \frac{\delta_i}{\lambda_i} \implies k_{\text{tot}} = \frac{1}{k_{\text{tot}}^{-1}}$$

## 3. Heat conductivity through a hollow cylinder

$$\dot{Q} = -\lambda A \frac{\mathrm{d}T}{\mathrm{d}r} = -\lambda 2\pi r l \frac{\mathrm{d}T}{\mathrm{d}r}$$

$$\boxed{\dot{Q} = 2\pi l \cdot \frac{T_0 - T_1}{\frac{1}{\lambda} \ln \frac{r_1}{r_0}} = 2\pi r_0 l k \Delta T}$$

With multilayered walls:

$$\dot{Q} = 2\pi l \cdot \frac{T_0 - T_n}{\sum_{i=1}^{n} \frac{1}{\lambda_i} \ln \frac{r_i}{r_{i-1}}} = \boxed{2\pi r_0 l k_0 (T_0 - T_n)}$$

### 3.1 Thermal conductivity $k$

$$k_0 = \frac{1}{r_0 \left( \left(\frac{1}{\lambda_1} \ln \frac{r_1}{r_0}\right) + \left(\frac{1}{\lambda_2} \ln \frac{r_2}{r_1}\right) + \cdots + \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}}\right)\right)}$$

$$k_0^{-1} = r_0 \sum_n \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}}\right)$$

### 3.2 Thin-walled pipe approximated as a flat wall

$$\dot{Q}_T = \lambda A \frac{T_0 - T_1}{\delta} \quad ; \quad A = 2\pi l \left(r_0 + \frac{\delta}{2}\right)$$

$$\frac{\dot{Q}_T}{\dot{Q}_W} = \frac{\delta}{\left(r_0 + \frac{\delta}{2}\right) \ln \left(1 + \frac{\delta}{r_0}\right)} = \frac{\delta}{r_0} \frac{1}{\left(1 + \frac{\delta}{2r_0}\right) \ln \left(1 + \frac{\delta}{r_0}\right)}$$

## 4. Convective heat transfer

### 4.1 Reynold's number

It compares inertial to viscous forces.

$$\boxed{Re = \rho\, c \frac{L_{\text{char}}}{\eta} = c \frac{L_{\text{char}}}{\nu}}$$

$Re < 2300$: Laminar flow;
$Re \approx 2300$: Transition zone;
$Re > 2300$: Turbulent flow

### 4.2 Prandtl number

It compares momemtum diffusivity to thermal diffusivity.
Sets velocity's relative thickness and thermal boundary layers.

$$Pr = \frac{\nu}{a} = \frac{\delta_{\text{fluid}}}{\delta_{\text{th}}} = \frac{\eta c_p}{\lambda} \quad ; \quad a = \frac{\lambda}{\rho c_p} \quad ; \quad \nu = \frac{\eta}{\rho}$$

### 4.3 Nusselt number

Measures convective heat transfer relative to pure conduction across a boundary layer and the heat transfer coefficient $\alpha$.

$$Nu = \frac{\alpha \cdot L_{\text{char}}}{\lambda_{\text{fl}}} = \dot{q} \frac{L_{\text{char}}}{\lambda \Delta T_{\text{char}}}$$

$$\alpha = \frac{Nu \cdot \lambda}{L_{\text{char}}}$$

#### 4.3.1 Forced convection

$$\boxed{Nu = c \cdot Re^m \cdot Pr^n}$$

where $c$ is a constant.

For laminar flow over a flat plate:

$$Nu = 0.664 \cdot \sqrt{Re} \cdot \sqrt[3]{Pr}$$

For a flow in the transition zone (Tz):

$$Nu_{\text{Tz}} = \sqrt{Nu_{\text{lam}}^2 + Nu_{\text{turb}}^2}$$

### 4.4 Grashof number

It compares buoyancy to viscous forces.

$$Gr = g\beta\Delta T \frac{L^3}{\nu^2} \quad ; \quad g = 9.81 \text{ N/kg}$$

$$\beta\Delta T = \frac{\rho_\infty - \rho_0}{\rho_\infty} = \frac{\Delta\rho}{\rho_\infty}$$

$$Gr = g \frac{\rho_\infty - \rho_0}{\rho_\infty} \frac{L^3}{\nu^2}$$

#### 4.4.1 Natural convection

**Peclet number**

It compares advective to diffusive heat transport, i.e., the strength of convection relative to thermal diffusion.

$$Pe = RePr$$

**Rayleigh number**

It combines buoyancy driving and thermal diffusion, and it measures the intensity of natural convection.

$$Ra = GrPr$$

## 5. Heat exchanger

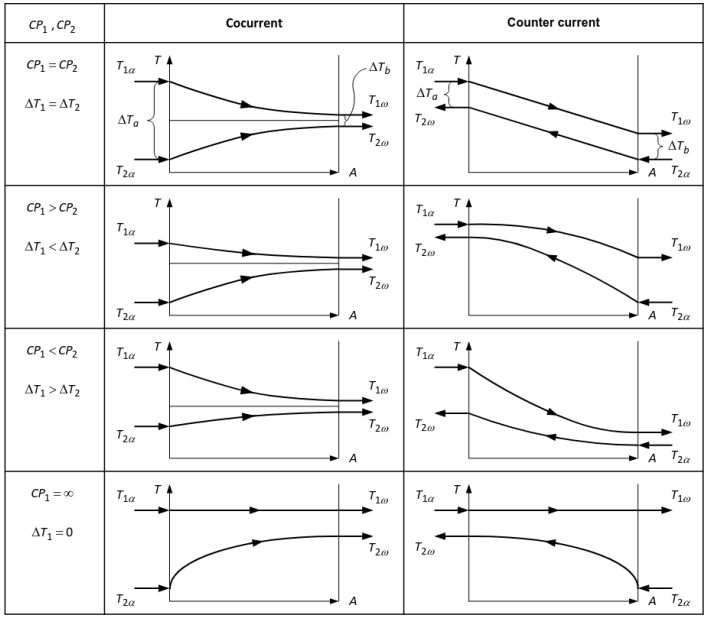
### 5.1 Energy balance

$$\dot{Q} = \dot{m} c_p \Delta\vartheta = CP\Delta\vartheta$$

$$\dot{Q}_1 = \dot{Q}_2$$

$$\dot{m}_1 c_{p1} \Delta\vartheta_1 = \dot{m}_2 c_{p2} \Delta\vartheta_2 \Longleftrightarrow CP_1 \Delta\vartheta_1 = CP_2 \Delta\vartheta_2$$

### 5.2 Average temperature difference




#### 5.2.1 Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \dfrac{\Delta T_a}{\Delta T_b}}$$

where:

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

#### 5.2.2 Mean temperature heat flow

$$\dot{Q} = kA\Delta T_m$$

#### 5.2.3 Heat capacity rate ratio (Capacity ration)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$R_1 = \frac{\Delta\vartheta_1}{\Delta\vartheta_2} = \frac{CP_1}{CP_2}$$

### 5.3 Cocurrent vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{\text{co}}\Delta T_{m,\text{co}} = kA_{\text{count}}\Delta T_{m,\text{count}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}$$

#### 5.3.1 If $\dot{Q}_1 \neq \dot{Q}_2$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\dot{Q}_{\text{count}}\Delta T_{m,\text{count}}}{\dot{Q}_{\text{co}}\Delta T_{m,\text{co}}}$$

# II. The 2nd law of thermodynamics

## 6. Recap of the 1st law

### 6.1 Closed systems

$$q_{12} + w_{12} = u_2 - u_1$$

### 6.2 Open systems

$$q_{12} + w_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} = \int v\,dp + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right) + j_{12}$$

$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 \overset{\text{ideal gas}}{=\!=\!=} c_p\left(T_2 - T_1\right)$$

## 7. The 2nd law

### 7.1 Statements

$$\boxed{\text{Energy} = \text{Exergy} + \text{Anergy}}$$

| Processes | Reversible | Irreversible |
|---|---|---|
| **Energy** | Conserved | Reduced → Loss |
| **Entropy** | Constant | Increases → Production |

### 7.2 Entropy

Entropy $S$: extensive state variable [J/K]

Specific entropy $s$: intensive state variable [J/kgK]

Entropy flow $\dot{S}$: entropy over time [W/K]

## 8. Entropy balance equation

$$\boxed{dS = dS_Q + dS_m + dS_{irr}}$$

### 8.1 Entropy balance terms

- Heat transfer across the system boundary:

$$dS_Q = \frac{dQ}{T}$$

- Mass flow across the system boundary:

$$dS_{\text{m}} = \dot{S} = \dot{m}\,s$$

- Irreversible processes inside the system:

$$\dot{S}_{irr}(t)\begin{cases} dS_{irr} > 0 & \text{irreversible processes} \\ dS_{irr} = 0 & \text{reversible processes} \\ dS_{irr} < 0 & \text{impossible} \end{cases}$$

### 8.2 Entropy balance equation for closed systems

$$dS = dS_Q + \cancelto{0}{dS_m} + dS_{irr}$$

$$dS = \dot{S}_Q\,dt + \dot{S}_{irr}\,dt = \left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right)dt$$

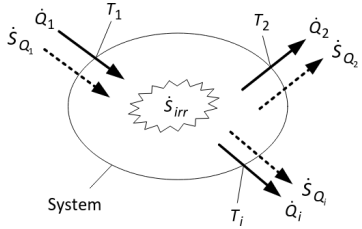
#### 8.2.1 Closed system entropy flow balance equation

$$\frac{dS}{dt} = \sum_i \dot{S}_{Qi} + \dot{S}_{irr} \quad ; \quad \dot{S}_{irr} \geq 0$$

For steady-state case:

$$\dot{S}_{irr} = -\sum_i \dot{S}_{Qi} = -\sum_i \frac{\dot{Q}_i}{T_i} \geq 0$$

#### 8.2.2 Visual representation




$$\boxed{\sum_i \dot{S}_{Qi} = \sum_i \frac{\dot{Q}_i}{T_i} = -\dot{S}_{irr}}$$

### 8.3 Entropy balance equation for open systems

$$\frac{dS}{dt} = \sum_\alpha \dot{m}_\alpha s_\alpha - \sum_\omega \dot{m}_\omega s_\omega + \sum \dot{S}_Q(t) + \dot{S}_{irr}(t)$$

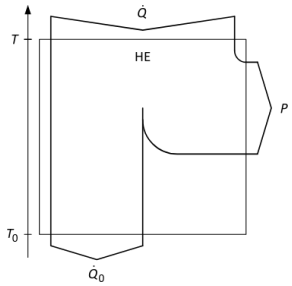
#### 8.3.1 Steady-state flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha - \sum \dot{S}_Q \geq 0$$

#### 8.3.2 Steady-state and adiabatic flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha \geq 0$$

## 9. The thermal enigne




$$\dot{Q} + \dot{Q}_0 + P = 0$$

$$\eta_{th} = \frac{-P}{\dot{Q}} = 1 - \frac{\left|\dot{Q}_0\right|}{\dot{Q}}$$

$$\dot{S}_{irr} = -\sum_{i=1}^{n} \dot{S}_{Qi} = -\frac{\dot{Q}}{T} - \frac{\dot{Q}_0}{T_0} \geq 0$$

$$-\dot{Q}_0 = T_0\left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right)$$

### 9.1 Carnot efficiency

$$\eta_{th} = 1 - \frac{T_0}{T} - \cancelto{0,\ \text{ideal}}{\frac{T_0\dot{S}_{irr}}{\dot{Q}}}$$

$$\eta_{th,max} = \eta_C(T, T_0) = 1 - \frac{T_0}{T}$$

# 10. Entropy $S$ as a state variable

## 10.1 Reversible processes

### 10.1.1 Clausius entropy relation

$$\boxed{\delta q_{\text{rev}} = T\,ds \iff \delta Q_{\text{rev}} = T\,dS}$$

### 10.1.2 Work $W$ and heat $Q$

$$w_{p,\,rev} = \int v\,dp$$

$$Q_{rev} = \int T\,dS$$

## 10.2 Dissipation $j$ as a state variable

### 10.2.1 Dissipation

$$w_{t12} = w_{p12} + j_{12} \iff w_{p12} = w_{t12} - j_{12}$$

$$dq_{rev} + dw_{v,\,rev} = du$$

$$T\,ds = dq + T\,ds_{irr} = dq_{rev} + dj$$

$$\int_1^2 T\,ds = q_{12} + j_{12}$$

### 10.2.2 For closed systems

$$ds = \frac{du + p\,dv}{T} = \frac{d_v\,dT}{T} + \frac{R_i\,dv}{v}$$

$$s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1}$$

### 10.2.3 Work change

All the works can be calculated with a temperature change:

$$q_{12} + w_{t12} = h_2 - h_1 + \frac{c_2^2 - c_1^2}{2} + g\,(z_2 - z_1)$$

$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 = c_p\,(T_2 - T_1)$$

$$\int T\,ds = q_{12} + j_{12} = \boxed{\int T\,ds + \int v\,dp = h_2 - h_1}$$

$$\boxed{T\,ds + v\,dp = h_2 - h_1 \iff T\,ds - p\,dv = u_2 - u_1}$$

## 10.3 Reversible state changes with perfect gases

### 10.3.1 Open system

$$q_{12} + j_{12} + w_{p,rev} = h_2 - h_1$$

$$\int_1^2 T\,ds + \int_1^2 v\,dp = c_p\,(T_2 - T_1)$$

### 10.3.2 Closed system

$$q_{12} + j_{12} + w_{p,rev} = u_2 - u_1$$

$$\int_1^2 T\,ds - \int_1^2 p\,dv = c_v\,(T_2 - T_1)$$

## 10.4 Polytropic state changes

$$T\,ds = dq + dj = du + p\,dv = dh - v\,dp$$

$$du + p\,dv = c_v\,dT + p\,dv \quad ; \quad dh - v\,dp = c_p\,dT - v\,dp$$

$$dw_v = c_v\frac{k-1}{n-1}\,dT = -p\,dv$$

For the change in entropy of the polytropic state change:

$$ds = c_v\frac{n-k}{n-1}\frac{dT}{T}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$

$$\boxed{s_2 - s_1 = c_v\frac{n-k}{n-1}\ln\frac{T_2}{T_1}} \quad n \neq 1$$

$$\boxed{s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1} = c_p \ln\frac{T_2}{T_1} - R_i \ln\frac{p_2}{p_1}}$$

**Compressor with dissipation**: $n > k$
**Turbine with dissipation**: $n < k$

# 11. Exergy and anergy

**Exergy**: useful part of energy that can be completely converted into work under given ambient conditions.

**Anergy**: remaining part of energy that cannot be converted into work under given ambient conditions.

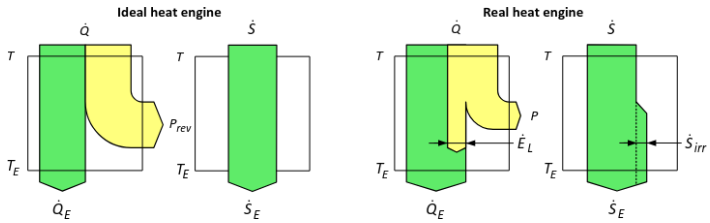
$$\boxed{e = ex + an}$$

## 11.1 Statements of the 2nd LT

**I.** In all irreversible processes exergy is converted in anergy
**II.** Only in reversible processes exergy remains constant
**III.** It is impossible to convert anergy into exergy

## 11.2 Exergy

### 11.2.1 Exergy loss




$$\boxed{\dot{E}_L = T_E\dot{S}_{\text{irr}} = T_E\,\dot{m}\,(s_2 - s_1) = T_E\underbrace{\frac{\Delta T}{T_1 \cdot T_2}\dot{Q}}_{\dot{S}_{\text{irr}}}}$$

### 11.2.2 Exergy of heat

$$\dot{E}_Q = P_{rev} = \left(1 - \frac{T_E}{T}\right)\dot{Q} = \eta_C\dot{Q}$$

### 11.2.3 Exergy of material flows (without mixtures)

$$e = h - h_E - T_E\,(s - s_E) + \frac{c^2}{2} + gz + q\left(1 - \frac{T_E}{T}\right)$$

$$e_2 - e_1 = h_2 - h_1 - T_E\,(s_2 - s_1)$$

**Ideal gases**

$$e = c_p\left(T - T_E - T_E\ln\frac{T}{T_E}\right) + R_iT_E\ln\frac{p}{p_E}$$

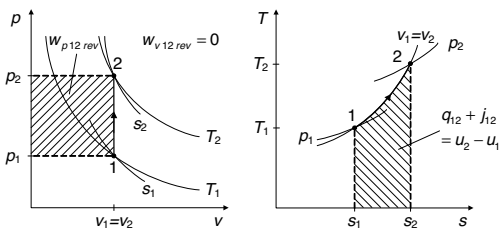
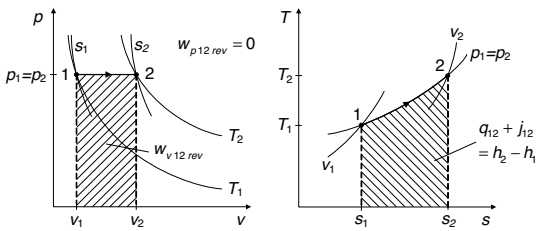
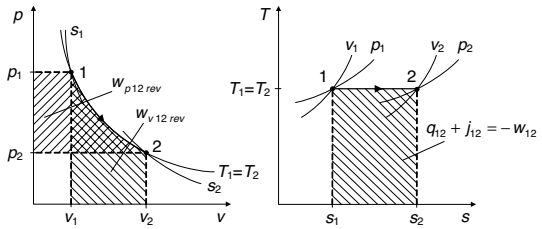
**Isobaric**

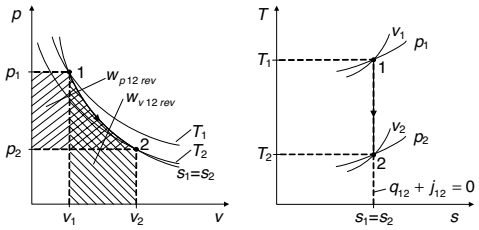
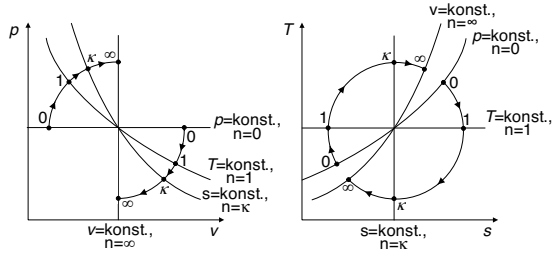
$$e = c_p\left(T - T_E - T_E\ln\frac{T}{T_E}\right)$$

**Isobaric for $T < T_E$ (cooling)**

$$e = c_p\left(T_E - T - T_E\ln\frac{T_E}{T}\right)$$

## Special state changes with perfect gases (without consideration of kinetic and potential energies)

| **Ideal gas** | $v =$ **constant: Isochoric state change** | $p =$ **constant: Isobaric state change** | $T =$ **constant: Isothermal state change** |
|---|---|---|---|
| **Therm. state equation**<br>$p\,V = n\,R\,T$<br>$R = 8.314\,\mathrm{J/mol\,K}$<br>$p\,V = m\,R_i\,T$<br>$R_i = \dfrac{R}{M_i}$<br>$p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$<br>$R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$<br>$c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$ |  <br>$v = \text{const.}: \quad \dfrac{p_1}{T_1} = \dfrac{p_2}{T_2} = \dfrac{p}{T} = \text{const.}$<br>$w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$<br>$w_{p\,12\,rev} = v\,(p_2 - p_1) = \dfrac{p_2 - p_1}{\rho}$ |  <br>$p = \text{const.}: \quad \dfrac{v_1}{T_1} = \dfrac{v_2}{T_2} = \dfrac{v}{T} = \text{const.}$<br>$w_{v\,12\,rev} = -p\,(v_2 - v_1) = -R_i\,(T_2 - T_1)$<br>$w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$ |  <br>$T = \text{const.}: \quad p_1\,v_1 = p_2\,v_2 = p\,v = \text{const.}$<br>$w_{v\,12\,rev} = -p_1\,v_1\,\ln\dfrac{v_2}{v_1} = \underbrace{p_1\,v_1\,\ln\dfrac{p_2}{p_1}}_{[\mathrm{J}]} = \underbrace{R_i\,T\,\ln\dfrac{p_2}{p_1}}_{[\mathrm{J/kg}]}$<br>$w_{p\,12\,rev} = w_{v\,12\,rev}$ |
| **1st LT closed system**<br>$q_{12} + w_{12} = u_2 - u_1$<br>Rev. volume change work:<br>$w_{v\,12\,rev} = -\int_1^2 p\,\mathrm{d}v$<br>$w_{v\,12} = w_{v\,12\,rev} + j_{12}$<br>Overall work:<br>$w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$ | $q_{12} + w_{12} = u_2 - u_1$<br>Rev. SC:<br>$w_{12} = w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$<br>$q_{12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$ | $q_{12} + w_{12} = u_2 - u_1$<br>Rev. SC:<br>$w_{12} = w_{v\,12\,rev} = -p\,(v_2 - v_1)$<br>$q_{12\,rev} = u_2 - u_1 + p\,(v_2 - v_1) = h_2 - h_1$<br>$q_{12\,rev} = c_p\,(T_2 - T_1)$ | $q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1) = 0$<br>$q_{12} = -w_{12}$<br>Rev. SC:<br>$q_{12\,rev} = -w_{v\,12\,rev}$<br>$q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$ |
| **1st LT open system**<br>$q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. pressure change work:<br>$w_{p\,12\,rev} = \int_1^2 v\,\mathrm{d}p$<br>Technical work:<br>$w_{t\,12} = \int_1^2 v\,\mathrm{d}p + j_{12}$ | $q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. SC:<br>$w_{t\,12} = w_{p\,12\,rev} = v\,(p_2 - p_1)$<br>$q_{12\,rev} = h_2 - h_1 - v\,(p_2 - p_1) = u_2 - u_1$<br>$q_{12\,rev} = c_v\,(T_2 - T_1)$ | $q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. SC:<br>$w_{t\,12} = w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$<br>$q_{12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$ | $q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1) = 0$<br>$q_{12} = -w_{t\,12}$<br>Rev. SC:<br>$q_{12\,rev} = -w_{p\,12\,rev}$<br>$q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$ |
| **2nd LT / entropy**<br>$\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$<br>$\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$ | $\mathrm{d}s = \dfrac{\mathrm{d}u}{T} = \dfrac{c_v\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1}$ | $\mathrm{d}s = \dfrac{\mathrm{d}h}{T} = \dfrac{c_p\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_p\,\ln\dfrac{T_2}{T_1}$ | $\mathrm{d}s = \dfrac{p\,\mathrm{d}v}{T} = \dfrac{R_i\,\mathrm{d}v}{v} \Rightarrow s_2 - s_1 = R_i\,\ln\dfrac{v_2}{v_1}$<br>$\mathrm{d}s = -\dfrac{v\,\mathrm{d}p}{T} = -\dfrac{R_i\,\mathrm{d}p}{p} \Rightarrow s_2 - s_1 = -R_i\,\ln\dfrac{p_2}{p_1}$ |

| **Ideal gas** | $s =$ **constant: Isentropic state change** | $n =$ **constant: Polytropic state change** |
|---|---|---|
| **Therm. state change**<br>$p\,V = n\,R\,T$<br>$R = 8.314\,\mathrm{J/mol\,K}$<br>$p\,V = m\,R_i\,T$<br>$R_i = \dfrac{R}{M_i}$<br>$p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$<br>$R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$<br>$c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$ |  <br><br>$s = \text{const}: \quad p_1\,v_1^\kappa = p_2\,v_2^\kappa = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} = \left(\dfrac{v_1}{v_2}\right)^{\kappa-1}$<br>$w_{v\,12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$<br>$w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$ |  <br><br>Polytrope: $pv^n = \text{konst.}$<br>Isobare: n=0, $pv^0 = \text{konst.}$<br>Isotherme: n=1, pv = konst.<br>Isentrope: n=κ, $pv^\kappa = \text{konst.}$<br>Isochore: n=∞, $pv^\infty = \text{konst.}$<br>$n = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{v_1}{v_2}} = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{p_2}{p_1} - \ln\frac{T_2}{T_1}}$<br>$n = \text{const}: \quad p_1\,v_1^n = p_2\,v_2^n = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{n-1}{n}} = \left(\dfrac{v_1}{v_2}\right)^{n-1}$<br>$w_{v\,12} = \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$<br>$w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$ |
| **1st LT closed system**<br>$q_{12} + w_{12} = u_2 - u_1$<br>Rev. volume change work:<br>$w_{v\,12\,rev} = -\int_1^2 p\,\mathrm{d}v$<br>$w_{v\,12} = w_{v\,12\,rev} + j_{12}$<br><br>Overall work:<br>$w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$ | $q_{12} + w_{12} = u_2 - u_1$<br>Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$):<br>$w_{12} = w_{v12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$<br>$w_{v12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$ | $q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1)$<br>$w_{12} = w_{v\,12} = \dfrac{R_i}{n - 1}\,(T_2 - T_1) = c_v\,\dfrac{\kappa - 1}{n - 1}\,(T_2 - T_1) =$<br>$= \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$<br>Rev. SC:<br>$w_{12} = w_{v12\,rev}$<br>$q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ |
| **1st LT open system**<br>$q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. pressure change work:<br>$w_{p\,12\,rev} = \int_1^2 v\,\mathrm{d}p$<br>Technical work:<br>$w_{t\,12} = \int_1^2 v\,\mathrm{d}p + j_{12}$ | $q_{12} + w_{t\,12} = h_2 - h_1$<br>Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$):<br>$w_{t\,12} = w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$<br>$w_{p\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \kappa\,\dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$ | $q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1)$<br>$w_{t\,12} = w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$<br>Rev. SC:<br>$w_{p\,12\,rev} = n\,w_{v\,12\,rev} = \dfrac{n\,R_i}{n - 1}\,(T_2 - T_1) = \text{etc.}$<br>$q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ |
| **2nd LT / entropy**<br>$\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$<br>$\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$ | $\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T} = 0 \Rightarrow s_2 - s_1 = 0$ | $s_2 - s_1 = c_v\,\dfrac{n - \kappa}{n - 1}\,\ln\dfrac{T_2}{T_1} \quad (\text{for } n \neq 1)$<br>$s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1} + R_i\,\ln\dfrac{v_2}{v_1} = c_p\,\ln\dfrac{T_2}{T_1} - R_i\,\ln\dfrac{p_2}{p_1}$ |